**University of Eloued**
**Academic Year: 2025/2026**

**Lesson: Linear Regression and Logistic Regression**

*Dr. Medini Atmane*
*University of Eloued*
*Academic Year: 2026/2027*

Short Summary and Series

# *Series 04*
# *Multiple Logistic Regression*

## Example 2 — Multiple Logistic Regression

### Using Python Coding Method

Multiple Logistic Regression extends simple logistic regression by using **more than one independent variable** to predict a binary dependent variable.

### Variables

**Independent variables:**

- `Study_Hours`
- `Attendance_Percent`
- `Assignments_Completed`
- `Previous_Score`
- `Sleep_Hours`

**Dependent variable:**

- `Pass` (`0 = Fail`, `1 = Pass`)

**Link :**

https://github.com/atman500/Logistic_Regression_Deep_Learning-2026-2027/blob/main/Simple_logistic_corrected.ipynb

### Section 1 — Model Specification & Theoretical Framework

1. Specify the multiple logistic regression model linking the binary dependent variable (`Pass`) to the five independent variables, and state its formulation in terms of both the log-odds and the logistic (sigmoid) response function.

2. Explain why Ordinary Least Squares (OLS) is theoretically inadequate for binary dependent variables and why Maximum Likelihood Estimation (MLE) via iterative numerical solutions (such as Newton-Raphson) is required.

## Section 2 — Empirical Estimation

Using Python (`statsmodels`):

1. Fit the multiple logistic regression model and extract the parameter estimates, standard errors, z-statistics, and p-values. **And** Present the estimated regression equation in both log-odds and probability formats using the empirical coefficients.
2. Interpret the estimated regression coefficient and the corresponding odds ratio for a key predictor (e.g., `Study_Hours`), explaining the exact multiplicative effect on the odds of passing.
3. Evaluate the statistical significance of the independent variables using their p-values and confidence intervals.
4. Explain the precise baseline meaning and managerial role of the intercept term in the estimated model.

## Section 4 — Prediction & Threshold Management

Consider a new student profile:

**Study_Hours:** 6

- ✓ **Attendance_Percent:** 85
- ✓ **Assignments_Completed:** 8
- ✓ **Previous_Score:** 70
- ✓ **Sleep_Hours:** 7

1. Calculate the value of the linear predictor (Z) and compute the student's exact predicted probability of passing.
2. Analyze how altering the classification threshold from the standard $0.5$ to alternative cutoffs (e.g., 0.45 or 0.75) impacts the trade-off between Type I and Type II errors.

## Section 5 — Performance Evaluation & Strategic Risk Analysis

Using the testing dataset results:

Construct the:

- ✓ confusion matrix and compute
- ✓ Accuracy, Precision, Recall,
- ✓ and the F1-score.

2. Critically evaluate each performance metric in the context of academic risk management.
3. Determine which metric is most critical if the university's strategic objective is to minimize undetected student failures (False Negatives), and justify your reasoning.

**Solution:**

You can find  the full Solution in the  next git_hub Link:

https://github.com/atman500/Logistic_Regression_Deep_Learning-2026-2027/blob/main/Mult_logist.ipynb